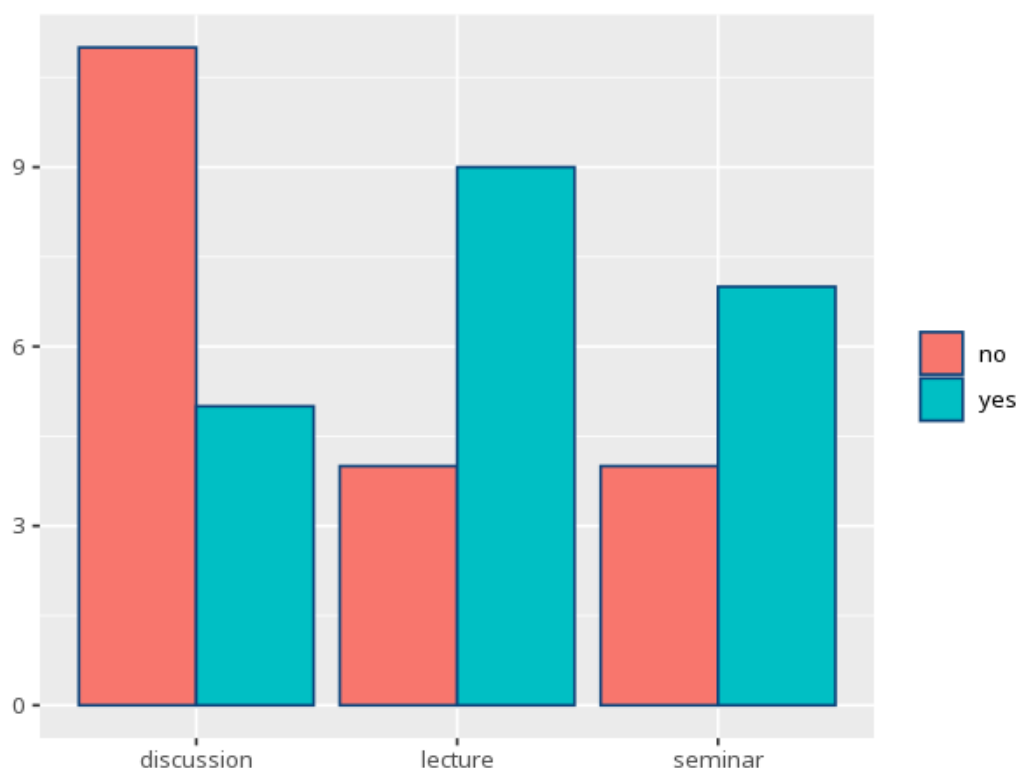


# 1 Chi-square independence

| Row \ Column | no                               | yes                              | Total |
|--------------|----------------------------------|----------------------------------|-------|
| discussion   | 11 [7.60] (68.8% / 57.9%) r=2.20 | 5 [8.40] (31.3% / 23.8%) r=-2.20 | 16    |
| lecture      | 4 [6.17] (30.8% / 21.1%) r=-1.47 | 9 [6.83] (69.2% / 42.9%) r=1.47  | 13    |
| seminar      | 4 [5.22] (36.4% / 21.1%) r=-0.87 | 7 [5.78] (63.6% / 33.3%) r=0.87  | 11    |
| Total        | 19                               | 21                               | 40    |

| $\chi^2$ | df | p    | Cramér's V [95% CI] |
|----------|----|------|---------------------|
| 4.90     | 2  | .086 | 0.35 [0.00, 1.00]   |



Tests whether two categorical variables are associated. A p below alpha means they are related; Cramér's V gives the strength of association (0–1); each cell's standardized residual (r) beyond about  $\pm 1.96$  flags a category combination that significantly drives the association. Valid only when expected counts are mostly  $\geq 5$  (the app flags this and suggests Fisher's exact) and each case appears once (raw counts, not percentages). Your significance threshold ( $\alpha$ ) is 0.05.

A chi-square test of independence gave  $\chi^2(2, N=40)=4.90$ ,  $p = .086$ ,  $V=.35$  [0.00, 1.00].

Statistical basis: Chi-square test of independence (Pearson, K. (1900). "On the criterion that a given system of deviations from the probable in the case of

a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling." *Philosophical Magazine*, 50(302), 157-175.); Cramér's  $V$  effect size (w-family; Cohen, 1988) (Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Routledge.).